CT Contrast Detection Using Deep Learning

Classifying CT slices as contrast-enhanced or non-contrast with CNNs on the Kaggle SIIM Medical Images dataset

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1. Problem and Motivation

IV iodinated contrast brightens vessels and enhancing tissue on CT. The **DICOM metadata that records contrast is often missing or wrong**: a large study found it missing or erroneous in 75.6% of 1,979 scans [1].

Our data shows the same: only **12 of 50 contrast slices** carry a ContrastBolusAgent entry.

Objective: predict contrast status from the pixels alone (binary classification).

2. Literature Review

- **Ye et al. [1]:** EfficientNetB4, 133,480 slices; AUC 0.996 internal, 1.0 external (head and neck).
- **FALCON:** ResNet9, 3,138 scans; external F1 98.1 to 99.4%.
- **Phase classification:** interclass relations [2], ResNet34, foundation-model embeddings.
- **Gap:** published models need large multi-centre data; we test **small-data feasibility** with controls and uncertainty.

3. Dataset

100	50/50	54	2
slices 512x512	contrast split	patient IDs	scanner makers

- DICOM + overview.csv; HU -3024 to 3072.
- Age 69.3 vs 65.7 (p = 0.093).
- 6 exact duplicate images; no corrupt files.

4. Methodology and Models

1. DICOM to HU
slope and intercept

2. Preprocess
window -160 to 240 HU, 224x224, ImageNet norm.

3. Augment
flip, 10 deg rotation, 5% shift, scale; *no intensity jitter*

4. Validate
5-fold stratified patient-grouped CV, fixed epochs

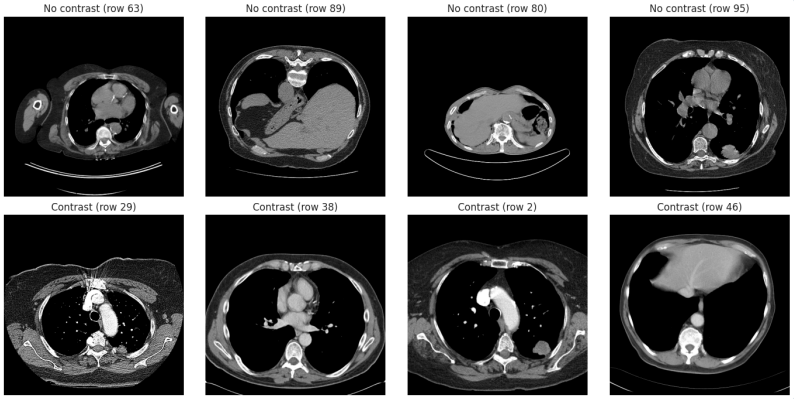
A. HU + LR: HU histogram, percentiles, mean, SD in body mask

B. Age only: confounding control

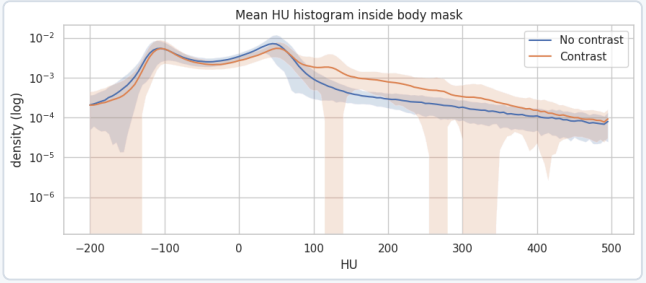
C. Small CNN: 4 conv blocks; 30 epochs, lr 1e-3

D. ResNet18: ImageNet, fine-tuned; 15 epochs, lr 1e-4

5. Exploratory Data Analysis



Top: no contrast. Bottom: contrast (bright vessels and heart chambers). Soft-tissue window.



Mean HU histogram in body mask: contrast shifts mass into 100-300 HU.

- Soft-tissue mean HU **38.2 vs 16.2** (p = 4e-7).
- 90th percentile **160.7 vs 83.7** (p = 5e-12).
- 99th percentile no difference (p = 0.49): a soft-tissue and vessel shift, not bone or air.

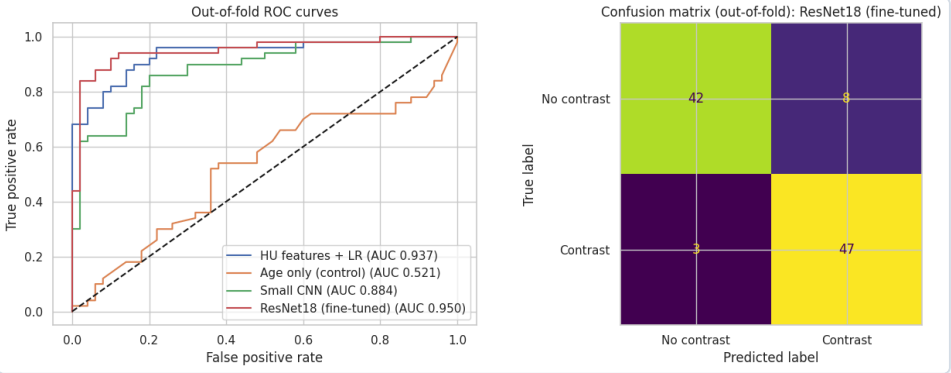
6. Results (5-fold CV)

Model	Acc	F1	Pooled AUC [95% CI]
A. HU + LR	0.862	0.852	0.937 [0.88, 0.98]
B. Age only	0.577	0.516	0.521 [0.39, 0.65]
C. Small CNN	0.835	0.829	0.884 [0.82, 0.95]
D. ResNet18	**0.894**	**0.888**	**0.950 [0.90, 0.99]**

Pooled AUC: all 100 images predicted once out-of-fold; CI by patient-group bootstrap.

ResNet18: precision 0.87, recall 0.93 (47 of 50 contrast slices found).

7. ROC Curves and Confusion Matrix (out-of-fold)



Right: ResNet18 confusion matrix (42 TN, 8 FP, 3 FN, 47 TP). Only 3 of 50 contrast slices were missed.

8. Key Insights

- Contrast is detectable from pixels even with 100 slices.
- **Simple HU features rival the CNN** (AUC 0.937 vs 0.950); CIs overlap, so the CNN advantage is **not established**.
- Age-only AUC 0.52: age is not driving results.
- ResNet18 most stable (accuracy SD 0.04 vs 0.12 for small CNN).

9. Limitations and Next Steps

- Only 100 slices; test folds hold 15 to 23 images.
- 6 duplicates kept; unique IDs do not guarantee independence; anatomy or scanner shortcuts possible; no external validation.
- **Next:** de-duplicate, nested CV, more data, Grad-CAM check, contrast-phase classification.

Conclusion

A fine-tuned ResNet18 detected IV contrast with pooled AUC 0.95 and 89% accuracy under patient-grouped CV, and simple HU features performed nearly as well. With 100 slices this is a proof of concept, not evidence of clinical readiness; external validation is required.

Data and code: Kaggle kmader/siim-medical-images; PyTorch on Google Colab.

References: [1] Ye Z et al., Deep learning-based detection of IV contrast enhancement on CT scans, Radiology: AI, 2022. [2] Muhamedrahimov R et al., Learning interclass relations for IV contrast phase classification in CT, MIDL 2021. FALCON open-source tool (Fintelmann Lab). He K et al., ResNet, CVPR 2016.

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